

A PROJECT REPORT
On
G+14 RESIDENTIAL BUILDING CONSTRUCTION

A PROJECT REPORT

Submitted by

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(200170106054)

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

In

B.E. - 4th YEAR, SEMESTER- VII

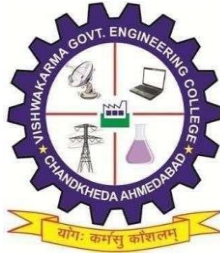
CIVIL ENGINEERING DEPARTMENT

Of

**VISHWAKARMA GOVERNMENT ENGINEERING
COLLEGE**

Gujarat Technological University, Ahmedabad

(October, 2023)



Vishwakarma Government Engineering College Chandkheda, 382424

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **G+14 Residential Building Construction** has been carried out by Jeelkumar Chaudhari under my guidance in partial fulfillment for the degree of Bachelor of Engineering in civil, 7th Semester of Gujarat technological university, Ahmadabad during the academic year 2023-2024.

Sign
Prof. Ketan L. Timani
(Internal guide)

Sign
Prof. N.P.SINGH
(Head of Department)

**Vishwakarma Government Engineering College
Chandkheda, 382424**

DECLARATION

we hereby declare that the internship/ project submitted along with the internship/project entitled **G+14 Residential Building Construction** submitted in partial fulfillment for the degree of Bachelor of Engineering in civil to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me at **Vishwakarma Government Engineering College, Chandkheda** under the supervision of **prof. Ketan L. Timani** and that no part of this report has been directly from any student's reports or taken from any other source, without providing due reference.

Name of the Student

Jeelkumar Chaudhari.

Sign of the student

COMPLETION CERTIFICATE

Shikshapatri™ Group

sky PARK
— 2-3 BHK corner living —

To whom so ever It may concern

This is to certify that. Mr.Jeelkumar Chaudhari (Enrollment number : 200170106054) of B.E civil engineering sem – VII, Vishwakarma Government Engineering college, Ahmedabad has Successfully Completed his 15 days internship about G+14 Residential building (From 27 July 2023 to 10 August 2023) at the Shikshapatri Group Gandhinagar.

He has undertaken his training at the Shikshapatri Group Gandhinagar with action Participation in execution work. He has learned the construction process, estimation of various components of structure and developed multiple skill including communication with labors and engineers, Supervision of sites and teamwork.

He was highly competent towards his work and his approach towards the training program was very positive. During this training period, we found him very sincere, hardworking, proactive and efficient.

We wish him great success in his Professional and Personal life.

MentorSignature

A. A. Chaudhari
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ACKNOWLEDGEMENT

I wish to express our sincere gratitude to our external guide Mr. Ketul Patel for continuously guiding me at the site and answering all my doubts with patience. I would also like to thank my internal guide Prof. Ketan L. Timani and head of the department prof. N. P. Singh for helping us through our internship by giving us the necessary suggestions and advises along with their valuable co-ordination in completing this internship.

The internship opportunity I had with Standard Infrastructure was a great chance for learning and professional knowledge development. Therefore, an opportunity provided to me may be a part of it. I am also very grateful for having a chance to meet so many wonderful, knowledgeable people and professionals who led me through this internship period. First, I thanks to Gujarat technological university to give us this opportunity in summer internship to gain practical knowledge of our stream I am very thankful to my internal guide prof. Ketan L. Timani who help me during internship and report preparation.

I also thanks my parent, friends and all the members of the family for their precious support and encouragement which they had provided in completion of our work. In addition to that, I would also like to mention the company personals who gave us the permission to use and experience the valuable resources required for the internship.

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ABSTRACT

In civil engineering, practical knowledge is important. While learning the basics in college is necessary, what truly matters is how well engineers can use that knowledge in real-life situations. Civil engineers face complex challenges while dealing with specific site such as limited budgets, rules and regulations, environmental aspects, etc. all while trying to come up with smart, effective and economical solutions. It's all about performing experience, being good at solving problems, and staying up-to-date with new technologies and industry methods. By combining theoretical knowledge and practical experience, civil engineers make structures like buildings and bridges that make our modern world possible.

The residential building construction modeling is carried out using Auto cad software and the analysis and design are carried out by STAAD pro software, In this study G+14 storey residential building construction is done and study the variation components of building like slab, column, beam, stair case etc. with its reinforcement detailing. Also studied the plan and elevation of the building as per architecture design, all design of various components studied are practically and theoretically as per Indian Standard and building bye laws.

This study works as great engineering as well as management skills development for students by knowing information such as assigning labels for particular work, procurement of materials well in advance, incentives of labour, site execution work, planning and precision as a civil engineer.

LITERATURE REVIEW

1. Salehuddun (2011)

He focused on nonlinear geometric analysis to be compared with linear analysis. In this investigation, a six story 2-D steel outline structure with 24 m stature has been chosen to be glorified as tall building model. The model was broke down by utilizing SAP2000 basic investigation programming with the thought of geometric nonlinear impact. This investigation demonstrated that a steel outline with the thought of wind stack create more noteworthy influence an incentive when contrasted with the steel outline without wind stack.

2. Dhanavath Seva (2017)

There are a few techniques for investigation of various edges like kani's strategy, cantilever technique, entrance technique, matrix strategy. The present venture manages the investigation of a multi-storied private working of G+6 comprising of 5 lofts in each floor. The dead load & live loads are connected and the plan for pillars, sections, balance is acquired STAAD Pro with its new highlights outperformed its ancestors, and compotators with its information offering abilities to other real programming like Auto-CAD, and MS Excel. We reason that Staad professional is an effective instrument which can spare much time and is extremely precise in Designs. Accordingly it is reasoned that staad expert bundle is appropriate for the plan of a multi storied building. The present task manages the investigation and outline of a multi storied private working of (ung-2+g+10) by utilizing most practical section strategy. The dead load & live loads are connected and the outline for shafts, sections, balance is acquired etabs with its new highlights outperformed its antecedents, and compotators with its information sharing.our fundamental point is to finished a multi-story building is to guarantee that the structure is protected and temperate against all conceivable stacking conditions and to satisfy the capacity for which they have built **safety** prerequisites must be so the structure can fill it need with the keep up cost detailed arranging of the structure for the most part originates from a few examinations made by town organizers, speculators, clients, draftsmen and different specialists .on that, and an auxiliary designer has the primary effect on the general basic plan and a draftsman is engaged with tasteful details for the plan of the structure, the dead load, live load, **seismic** and wind stack are considered.

2. F.Biasioli

At the broad side there are three rows of columns (axes A, B, C (and D in levels -1 and -2)), in length there are six rows (axes 1 to 6). These columns support the slabs. Three different slab types (see Chapter 1) have been considered regarding the slabs: a) Flat slab with height $h = 21$ cm, directly supported by the columns b) Slab on beams, 2 spans, $h = 18$ cm, loads flow through beams into columns c) Slabs with embedded lighting elements $h = 23$ cm, loads flow through the coffered ceiling beams into supporting beams, which lead into the columns. Several multiple loads are present: the dead load of the construction, the interior (finishing, pavement, etc.) and the facade, service loads and two environmental loads – wind and snow.

3. S.V. Nagender

The GUI (or client) speaks with the STAAD examination motor through the STD input document. That information record is a content document comprising of a progression of charges which are executed consecutively. The charges contain either directions or information relating to examination and additionally outline. The STAAD input document can be made through a word processor or the GUI Modelling office. All in all, any content tool might be used to alter/make the STD input record. The GUI Modelling office makes the info record through an intelligent menu driven designs situated method.

CHAPTER:1 INTRODUCTION

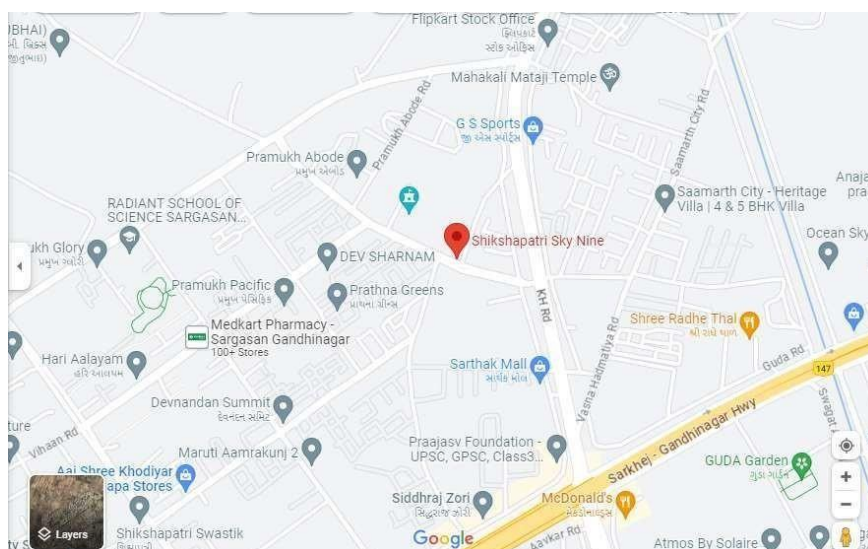
➤ 1.1 Project Information :-

- Name of project: Residential Building Construction
- Company name: Shikshapatri Group
- Location of site : Sargasan Gandhinagar
- No. of floors: Ground +14
- Project manager : Ketul Patel



(Figure 1)

➤ 1.2 Site Location :-



(Figure 2)

➤ I.3 Planning Of Building

1. Basement Plan :-



2. 2 & 3 BHK Plan :-



(Figure 3)

➤ 1.4 – Components Of Building

- Foundation
- Plinth
- Walls
- Column
- Sill
- Door
- Window
- Ventilator
- Slab
- Beam
- Lintel
- Stair
- Floors
- Parapet

❖ Foundation:-

The foundation of a building or civil structure is the element that connects it to the ground. It is located at the very bottom of the construction, in direct contact with the soil, and acts to transfer loads safely from the building to the ground. Foundations can be either shallow or deep, referring to the depth of the soil in which the foundation is built. For instance, most single – storey residential dwellings would have relatively shallow foundations, while an inner city skyscraper would require much deeper foundations to provide structural stability and support to the main building. Size of the foundation at the site – 5'*5'*5.

❖ Plinth:-

The portion of the building above the ground upto the top of the floor immediately above the ground is known as plinth. The plinth height should be such that after leveling of the ground adjoining of the building, the rain may be not enter the ground floor. The height of the plinth should not be less than 45cm. Damp proof course (DPC) is provided at the top of the plinth. Plinth protects the building from rain, damp or moisture and insects and transmitted load of the superstructure to the foundation. Size of the plinth level at the site – 450 mm.

❖ Column:-

A column is a structural element that transmits, through compression, the weight of the structure above to the other structural elements below. In other words, a column is called as a compression member. The column

which is a vertical member transfer loads from slab and beam directly to the subsequent soil. **Column** details (for this project):

Size of the **column**

Grade of **concrete** – M25

Grade of steel – Fe500

❖ **Door :-**

A door is a movable barrier secured in an opening, known as the doorway, through a building wall or partition wall. It provides access into the room, offers privacy of the sight and sound.

Size of the doors:

External door = 1.10*2.20 m

Internal door = 1.00*2.0 m

Bath and W.C. door = 0.80*2.0 m

❖ **Window:-**

A window is an opening made in the wall for providing light and ventilation.

Size of the window:

1.20*1.20 m

1.00*1.20 m

0.60*1.20 m

❖ **Stairs:-**

The stair is a series of steps arranged to connect the different floors of a building. The stair should be so located as to provide easy, safe and comfortable mode of communication. It may be made of timber, stone, bricks, steel or reinforced **concrete**, etc. In this project reinforced **concrete** stairs is constructed.

❖ **Roof:-**

A roof is the uppermost part of a building to cover the space below and protect it from sun, wind, rain and snow, etc. A roof may be a flat, pitched or curved in shape. The flat roof may be of RCC slab (10 to 20 cm thick) and pitched roof may be trussed structure covered with roof covering material or curved roof may be a shell or dome.

❖ **Floor:-**

The purpose of a floor is to provide a level surface for the occupants of building, furniture, equipment, etc. Flooring may consists of plain cement **concrete** base (PCC) 1:4:8 of 100 mm, finished with cement mortar (1:3) of 20 mm thick or of mosaic tiles or marble or Polished kotah stone, etc.

❖ Beam:-

A structural member which carries lateral or transverse force is termed as beam, Beams are the media by which all loads of slab transferred to the vertical supports of a structure.

❖ Lintel:-

A lintel is a horizontal member which is placed across the opening of a door or window. The bearing of lintel should be the minimum of 10 cm at both the ends

➤ 1.5 Material Used For Construction:-

● Sand

Sand is a generally granular material composed of finely divide rock and mineral particles. It is also defined based on its size, being finer than gravel and coarser than silt. It is one of the essential type of building materials. The most common content of the sand is silica. It is usually in the form of quartz. The second most common content is calcium carbonate. Sand is used to make concrete in demand nowadays.

Sources:- River, Desert Beach, Marine or ocean sand



(Figure 4)

● Cement

Cement is a binding agent for construction materials. Cement is produced by burning at high temperatures in a define mixture of calcareous, siliceous, and aluminous raw materials and crushing the resulting clinkers to a fine powder. When cement is mixed with water, a chemical reaction starts, and the powder transforms into a paste that binds all the materials with it. The properties of cement depends on the chemical composition, manufacturing

process, and degree of fineness. There are almost 16 types of cement used in construction, depending upon the type of structure. The most common types of cement used are Ordinary Portland Cement (OPC) and Portland Pozzolana cement (PPC).



(Figure 5)

Bricks:-

Bricks are manufactured by mounding clay in rectangular blocks of uniform predetermine size, drying them and then burning them in in a kiln. Standard size of brick – 190mm*90mm*90mm.



(Figure 6)

- **Aggregate:-**

Aggregate are the most commonly used material in construction. Aggregate is the main component of composite materials such as concrete. It gives strength to the overall composite material.



(Figure 6)

- **Reinforcement:-**

Steel reinforcement are steel bars that are provided in combination with plain cement concrete to make it reinforced concrete. Tensile property for concrete structures is obtained by incorporating steel reinforcement. The steel reinforcement is strong in both tension and compression. The tensile property provided by the steel reinforcement will prevent and minimize concrete cracks under tension loads.



(Figure 7)

➤ 1.6 Instruments Used In Construction :-

● Needel vibrator

Needle vibrators are used to eliminate air pockets that can remain when pouring concrete that can ruin the integrity of concrete. Also it increases the density and strength of concrete. Needle Vibrators are most commonly used vibrator for concrete. It consists of a steel tube (with one end closed and rounded) having an eccentric vibrating element inside it. The steel tube called poker is connected to an electric motor or a diesel engine through flexible tube.



Figure 8

● Grinding Machine

Portable grinders are handheld power tools that are used for grinding, cutting or polishing. They are mainly used for cutting notches and are also highly efficient for smoothing and removal of unnecessary materials. Grinding machines remove material from the work piece by abrasion. Grinders are normally used to either smooth or cut hard surfaces and materials depending upon the type of accessory and material being used.

● Tower Crane

Fixed Tower Crane is used for bigger construction projects. These cranes have Maximum unsupported height: 265 feet (80 m) and a working radius of up to 70 meters. It is used to lower, lift, and move heavy or bulk construction materials around. The work of the tower crane is crucial. Construction companies use tower cranes to lift heavy materials such as steel and concrete.



(Figure 9)

➤ **RMC Truck :-**

These trucks mix or batch the ready mix on the job site. This type of truck can mix as much or as little amount of concrete as needed. This type of truck delivers concrete in a plastic state to the site. It available capacity in 6 cum. RMC, 7 cum. RMC & 8 cum. RMC for large concrete Mixers. Ready-mixed concrete is used in construction projects where the construction site is not willing, or is unable, to mix concrete on site.



(Figure 10)

➤ **Stationary Concrete Pump:-**

Stationary concrete pump is a tool used for transferring liquid concrete by pumping. The advantage of a stationary pump is that the pipeline can be long so that the actual pump can be very far away from where the concrete is being placed. It has pumping capacity up to $65 \text{ m}^3/\text{hr}$ and vertical pumping capability up to 180 meters. Usually, piston pumps are used, because they can produce hundreds of atmospheres of pressure.

➤ **Rebar Cutting Machine:-**

Steel bar cutting machine is used for rapidly cutting various gauge steel wire. These cutting machines enable users to slice steel wire, steel bar, brass wire, aluminium wire, polycarbonate fibre, PVC pipe and various other wires into very thin gauge sizes. There are various types of rebar cutters like electric, battery-powered, dual-power and manual. Rebar Cutter is a machine widely used in building and construction to cut bars made of steel and other heavy metals.



(Figure 11)

CHAPTER:-2-SITE ACTIVITIES

➤ 2.1 Framed Structure :-

● What is Framed Structure?

Framing means to give shape and support to a structure by combining components in building. Hence, a framed structure is a structure having the combination of beam, column and slab to resist the gravity and other later loads.

Advantages and disadvantages of Framed Structure :-

- 1) Strength & Durability.
 - 2) Easy Fabrication in Different Sizes.
 - 3) Fire Resistance.
 - 4) Pest & Insect Resistant.
 - 5) Moisture & Weather Resistance
- 1) High Maintenance Costs.
 - 2) Higher Initial Cost.
 - 3) Thermal Conductivity.
 - 4) Reduced Flexibility On-Site.
 - 5) Supporting Structures.

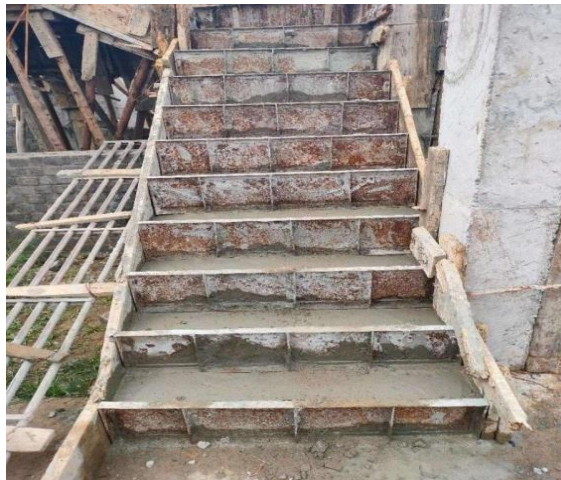


(Figure 12)

➤ 2.2 Staircase :-

1. Size of riser : First step = 225 mm and rest 150mm □
2. Size of tread = 300 mm □
3. No. of steps = 9 + landing □
4. Width of stair = 1.3 m □
5. Cover = 25 mm □

Here in staircase top and bottom have double layer steel. The top steel of landing is bent and taken as bottom steel in sloping slab and bottom steel of landing is taken as top in sloping slab so as to form scissor kachy shape. Spirit level meter is used for fixing stair formwork.



(Figure 13)

➤ 2.3 Beam Slab Casting :-

Casting a beam and slab for a building is a common construction process. Following are the general steps involved

- a. Formwork
- b. Reinforcement
- c. Pouring Concrete
- d. Curing
- e. Stripping Formwork
- f. Finishing

Shuttering of slab using wall plates of size 900*600*25mm. Beam reinforcement are placed as per detailing mentioned in drawing. 30 mm covers are tied in beam at bottom and side. Then white shuttering form sheet is spread on slab shuttering so as to get good finish at bottom. Over it slab beams are placed wherever required.

Concrete grade :M25

Grade of steel :Fe500



(Figure 14)

➤ 2.4 Plastering Work :-

Following are the some details about plastering work at a construction site :

Before plastering can begin, the walls and ceiling must be properly prepared. This often involves cleaning the surfaces, removing any loose or damaged plaster, applying a base coat or primer. Plaster can be made from various materials, including gypsum, cement, lime or a combination of these. The choice of plaster material depends on the specif requirements of the project.

Plaster is applied to the walls and ceilings using tools like trowel and plastering floats. Skilled plasterers use various techniques to achieve a smooth and even finish. Plastering is often done in multiple layers. The first layer is called the scratch coat, followed by one or more additional coats, depending on the desired finish. □ The drying time of plaster can vary depending on factors like humidity and temperature. It's important to allow sufficient time for each coat to dry before applying the next. After plastering is complete, the work area should be cleaned, and any debris or excess plaster should be properly disposed of.

➤ 2.5 Column :-

A column is a vertical structural member intended to transfer a compressive load. For example, a column might transfer loads from a ceiling, floor or roof slab or from a beam, to a floor or foundations.



(Figure 15)

➤ 2.6 Shuttering :-

● What is called shuttering?

Shuttering or formwork is the process of giving support and stability to the **concrete** before it becomes solid. The shuttering is commonly made using wood and steel. Shuttering is defined as the temporary mould that are used to hold the wet **concrete** in place until the **concrete** has set and cured.

● What is centering and shuttering?

Centering and shuttering are both formwork. Centering term is used when the formwork is horizontal like slab, floor beam, etc. Shuttering term is used when the formwork is vertical as in **column**, footing, retaining wall, etc.



(Figure 16)

CONCLUSION

This whole process of the study, from finding the site making to the final report getting it approved from the guide gave new experience at each and every step. Got the experience on the advancement in building construction, material management site planning, quality control, safety measures, quality estimates and some project management skills.

➤ The following things were learned:

- Increased technical skills.
- Explored the practical world outside the theoretical world.
- Improved communication skills.
- Got team working skills.
- Got experience on how to manage the labour.

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